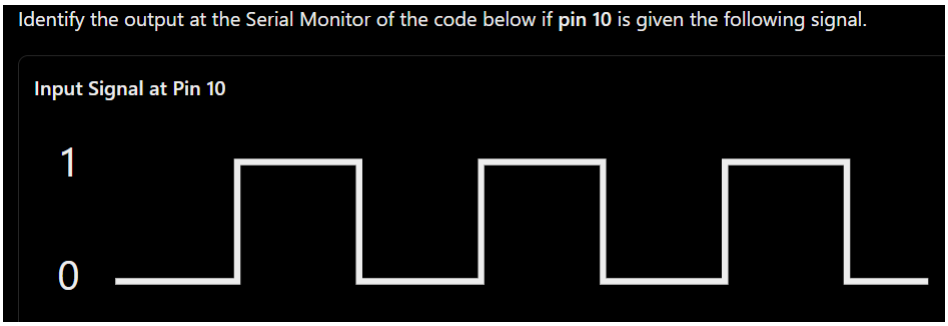
ASSIGNMENT-2

Arduino and Microcontroller Concepts (CSE-4820)

B.Tech. 7th Semester

1. Write an Arduino sketch to blink an LED 20 times, with a delay of 250 ms.
2. A system must detect both the movement or tilting of an object and the presence of a person nearby. Compare the working principles and limitations of an ultrasonic distance sensor and an optical Time-of-Flight (ToF) sensor, and evaluate whether a tilt sensor or a PIR sensor is more appropriate for detecting unauthorized movement of the object. Justify your sensor choices for the two detection tasks.
3. Write an Arduino sketch to convert a sensor reading from the range 0–1023 into a temperature range of 0–100 °C using the map() function and display the result on the Serial Monitor.
4. Differentiate between delay() and delayMicroseconds() in terms of the units of time they use and the situations in which each is appropriate. Then explain what millis() returns, how it can be used to track elapsed time, and why a program should account for its value eventually overflowing and returning to zero.
5. A pushbutton is connected between Arduino pin 2 and +5 V, with a 10 kΩ pull-down resistor between pin 2 and GND. State the digitalRead() value when the button is pressed and released, and explain why removing the resistor makes the input unreliable. Also differentiate between digitalRead() and analogRead(), and calculate the approximate analogRead() value for a 3.5 V input on A0 when the Arduino Uno uses a 5 V reference.
6. An Arduino reads an analog voltage at A0. Write a sketch to convert the ADC reading into voltage and display it on the Serial Monitor; if the measured voltage exceeds 3 V, display a warning. State the ADC resolution and the reference voltage assumption used in your conversion.
7. Write the sequence of outputs displayed on the Serial Monitor as the input signal at digital pin 10 changes between logic 0 and logic 1, as shown in the timing diagram.



```
int pin10;
void setup() {
    pinMode(10, INPUT);
    Serial.begin(9600);
}
void loop() {
    int ip = digitalRead(10);
    if (ip > 0) {
        Serial.println("HIGH");
    } else {
        Serial.println("LOW");
    }
}
```

8. Write an Arduino sketch that accepts a brightness value from 0 to 255 through the Serial Monitor and uses it to control an LED connected to PWM pin 9. Extend your explanation to distinguish the output of Serial.print("A"); Serial.print("B"); from Serial.println("A"); Serial.println("B");.
9. Demonstrate the purpose of pinMode() with a suitable Arduino example. Give an example of a string in Arduino C++ and demonstrate any two string functions, explaining what each function does.
10. Explain why overflow can occur in Arduino programs, describe one method of handling it with a suitable example, and list the commonly used Arduino C++ data types with a brief description of their purpose.
11. Write an Arduino sketch to read a potentiometer connected to A0 and turn ON an LED when the reading is greater than 500. Otherwise, turn it OFF.
12. An HC-SR04 ultrasonic sensor is connected to an Arduino with TRIG on digital pin 7 and ECHO on digital pin 8. Assume the speed of sound is 0.0343 cm/µs and that the ECHO pulse duration represents the sound's round-trip travel time. Write an Arduino sketch to generate the trigger pulse, measure the ECHO pulse duration, calculate the approximate distance in centimetres using distance = (duration × 0.0343) / 2, and display the result on the Serial Monitor. Also calculate the approximate distance for an ECHO pulse duration of 2000 µs, showing the steps.
13. Identify the output of the following Arduino code.

```
void setup() {
    Serial.begin(9600);
}
void loop() {
    int a = 22;
    int b = 13;
    int c = (a & b) | 2;
    Serial.println(c);
    while (1);
}
```

14. Develop an Arduino sketch that accepts commands through the Serial Monitor to control two appliances connected to pins 8 and 9. Support the commands L1ON, L1OFF, L2ON, and L2OFF. Explain how the program reads and interprets each command, and describe the purpose of Serial.begin(9600) and what the value 9600 represents.

15. Explain switch bounce and describe how software can determine that a switch state is genuine. Study the following debounce function, explain how it rejects false transitions, and state what change is needed if the switch may bounce for up to 30 ms. Discuss one drawback of this approach when the Arduino sketch must perform other tasks.

```
bool debounce(int pin) {
    bool state, previousState;
    previousState = digitalRead(pin);
    for (int counter = 0; counter < 10; counter++) {
        delay(1);
        state = digitalRead(pin);
        if (state != previousState) {
            counter = 0;
            previousState = state;
        }
    }
    return state;
}
```

16. Identify the output of the following Arduino code.

```
void setup() {
    Serial.begin(9600);
}
void loop() {
    int a = 15;
    int b = 10;
    int c = a & b;
    Serial.println(c);
    while (1);
}
```

17. Identify the output of the following Arduino code.

```
void setup() {
    Serial.begin(9600);
}
void loop() {
    int a = 40;
    int c = a >> 3;
    Serial.println(c);
    while (1);
}
```

18. State the logic levels obtained from a push-button connected to Arduino digital pin 7 using the internal pull-up resistor. Also, write the pin-configuration statement required for this interface.

19. Two temperature sensors are connected to analog pins A0 and A1. Write an Arduino code fragment that calculates the absolute difference between their readings and turns an LED ON if the difference exceeds 15. Briefly explain how the program processes the sensor readings and makes the decision.

20. Differentiate between digital and analog sensors with examples.

21. Explain the difference between local and global variables, and describe how array indexing works in Arduino. Use the sample sensor readings 120, 135, 142, and 128. Write a short sketch that declares a global array containing these four readings, accesses the first and last elements using the correct indices (0 and 3), and uses a local variable inside a function or block to calculate and print the sum of the first and last readings. Identify which variables are global and which are local.
22. Explain why Serial.available() should be checked before using Serial.read().
23. Explain the roles of void setup() and void loop() in an Arduino sketch. Then write a short Arduino UNO code snippet that demonstrates both digitalWrite() and analogWrite(), and explain the difference between their outputs.
24. A potentiometer is connected to analog pin A0 and an LED is connected to PWM pin 9. Write an Arduino sketch that reads the potentiometer value (0–1023), uses map() to convert it into an LED brightness value (0–255), and applies the result using analogWrite(). Calculate the mapped brightness for an input reading of 700, and explain the input and output ranges used by map() and why analogWrite() is used for the LED.
25. An Arduino must maintain the ON/OFF status of eight devices using a single byte and transmit a 16-bit integer as two separate bytes. Write a code fragment using bitSet(), bitClear(), and bitRead() to change and verify individual device states. Then use highByte() and lowByte() to split the integer value 258 (0x0102), and use word() to reconstruct the original value. Explain how the byte and bit-manipulation functions support these operations.